

Topology and resonances in a quasiperiodically forced oscillator

Brian K. Spears, Andrew J. Szeri*

Mechanical Engineering, University of California, Berkeley, CA 94720-1740, USA

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Abstract

In this paper we analyze resonance behavior in a quasiperiodically forced, nonlinear Mathieu equation. We develop a perturbation technique based on the method of multiple scales to find both a criterion for resonance and approximations to solutions in the neighborhood of a resonance. We compare the perturbation results to numerical solutions to validate both the resonance criterion and the approximate solutions. We also investigate the implications of resonance for the topology of attractors in the four-dimensional phase space. We show that a resonance occurs due to topological torus bifurcations (TTBs) and that resonant trajectories lie on topologically interesting knotted tori we have recently described elsewhere (Topological bifurcations of attracting 2-tori of quasiperiodically driven nonlinear oscillators, in review). The perturbation approximations capture both TTBs and the topology of invariant manifolds near resonance.

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1. Introduction

We study in this paper a quasiperiodic (QP) Mathieu equation with damping and cubic nonlinearities. The behavior of similar equations has been studied by a number of researchers in recent years. Stability and transition curves for a linear, undamped QP Mathieu equation were studied by Zounes and Rand [1] using a variety of methods including perturbation. A perturbation method was also employed by Alfred et al. [2] to study resonances in a QP linear Mathieu equation for the motions of a charged particle in a quadrupole ion trap. Zounes and Rand also investigated subharmonic resonances and chaos in a nonlinear QP Mathieu equation [3,4]. Guennoun and coworkers developed approaches for generating approximations to solutions of QP Mathieu equations with weak damping and

* Corresponding author. Tel.: +1 510 643 0298; fax: +1 510 642 6163.

E-mail address: aszeri@me.berkeley.edu (A.J. Szeri).

nonlinearity [5,6]. They apply the method of multiple scales twice in order to produce a doubly reduced system whose stationary solutions correspond to QP solutions of the original system. In work closely related to our multiple scales analysis which follows, Ramani et al. investigate bifurcations of periodic orbits in a quadratically damped Mathieu equation using a technique which perturbs off linear Mathieu equations [7].

We develop a multiple scale expansion technique to find the set of secondary frequencies ω_f that cause large amplitude solutions in the equation:

$$\ddot{z} + \mu \dot{z} + 4(\gamma + \alpha \cos 2t - \epsilon \cos 2\omega_f t)(-z + \chi z^3) = 0, \quad (1)$$

which models the axial motion of a charged particle in a quadrupole ion trap. By use of the multiple scale expansion, we are able to find the set of resonant frequencies, $\omega_{f,\text{res}}$, and the approximations to solutions for stability parameters γ and α lying in the first stable region for the linear Mathieu equation, including parts of the region where α is not small. In our multiple scales method, we solve a sequence of linear Mathieu equations order-by-order to find approximations of quasiperiodic solutions which lie on knotted 2-torus attractors.

Numerically produced response diagrams show large amplitude peaks located at $\omega_f = \omega_{f,\text{res}}$, i.e. at values which correspond to the resonant values predicted by the multiple scales expansion. By use of numerical simulations and the approximation scheme, we observe a sequence of bifurcations as ω_f is varied near the resonant value. Specifically, a large-amplitude knotted torus in the $\mathbb{R}^2 \times T^2$ phase space of the suspended system is generated and destroyed by a pair of *topological torus bifurcations* (TTBs) in a small neighborhood of the resonant ω_f value [8]. A TTB is a bifurcation through which a torus changes its knot type. We also discuss the topology of attracting tori, saddle-type tori, and their invariant manifolds as captured by both numerical solutions and multiple scales approximations.

2. Approximation of resonant solutions

2.1. Modified multiple scales technique

In order to examine the resonance behavior of (1), we introduce a modification of the method of multiple scales to find asymptotic solutions. We rescale the damping and nonlinearity in (1) so that it reads

$$\frac{d^2 z}{dt^2} + \epsilon \mu \frac{dz}{dt} + 4(\gamma + \alpha \cos 2t - \epsilon \cos 2\omega_f t)(-z + \epsilon \chi z^3) = 0 \quad (2)$$

with initial conditions:

$$z(0) = z^0, \quad \frac{dz}{dt}(0) = \dot{z}^0, \quad (3)$$

where $\epsilon \ll 1$ and χ , δ , and μ are $O(1)$ parameters. We also assume that the parameters α and γ are $O(1)$ and lie in the first stable region of the linear Mathieu stability diagram [9]. The result is a linear Mathieu equation modified slightly by damping, quasiperiodic excitation, and nonlinear terms. It is worth noting that traditional multiple scales analyses require α to be scaled by an $O(\epsilon)$ parameter, whereas we do not [10]. Also noteworthy is the fact that $O(\epsilon^0)$ parameters may take on values that appear small in magnitude relative to other terms in (2). The important issue, in the technical sense of the ordering symbol O , is how fast terms scale with ϵ . For example, motivated by a particular physical application, we will often choose γ to be small in magnitude. The ordering in terms of ϵ is preserved nonetheless, and our solution method can proceed without attending to the magnitude of γ .¹

We introduce an asymptotic expansion for the solution to (2):

$$z(\hat{t}, \tau) = z_0(\hat{t}, \tau) + \epsilon z_1(\hat{t}, \tau) + O(\epsilon^2), \quad (4)$$

¹ Ref. [10] contains a multiple scales calculation for a linear Mathieu equation where the parameter analogous to our γ may be arbitrarily small, but is still $O(\epsilon^0)$.

which evolves on the widely separated time scales $\hat{t} = t$ and $\tau = \epsilon t$. We remark that the two time scales $\hat{t}$ and τ are not the two frequencies of the QP forcing; that forcing happens only on the fast time scale $\hat{t}$. The slow time scale τ arises due to the cumulative effects of weak damping, nonlinearity, and the secondary forcing at ω_f . The total time derivative in terms of both the fast and slow time scales is

$$\frac{d}{dt} = \frac{\partial}{\partial \hat{t}} + \epsilon \frac{\partial}{\partial \tau}. \quad (5)$$

We substitute (4) and (5) into (2) and collect terms order-by-order in ϵ to obtain the sequence of problems:

$$O(\epsilon^0) : M(z_0) = \frac{\partial^2 z_0}{\partial \hat{t}^2} - 4(\gamma + \alpha \cos 2\hat{t})z_0 = 0, \quad (6)$$

$$O(\epsilon^1) : M(z_1) = 4\delta \cos(2\omega_f \hat{t})z_0 - 4\chi(\gamma + \alpha \cos 2\hat{t})z_0^3 - \mu \frac{\partial z_0}{\partial \hat{t}} - 2 \frac{\partial^2 z_0}{\partial \hat{t} \partial \tau} = 0. \quad (7)$$

For convenience we have defined the Mathieu operator $M \equiv \partial^2 / \partial \hat{t}^2 - 4(\gamma + \alpha \cos 2\hat{t})$.

At leading order, we have from [11] the solution for the homogeneous Mathieu equation, $M(z_0) = 0$, in the form:

$$z_0 = A \sum_{n=-\infty}^{\infty} D_{2n} \cos((2n + \beta)\hat{t}) + B \sum_{n=-\infty}^{\infty} D_{2n} \sin((2n + \beta)\hat{t}). \quad (8)$$

The fundamental frequency β and the coefficients D_{2n} depend on the stability parameters α and γ of the Mathieu equation. The fundamental frequency is given implicitly by the continued fraction expression:

$$\begin{aligned} \beta^2 = & -4\gamma + \frac{(2\alpha)^2}{(\beta + 2)^2 + 4\gamma - (2\alpha)^2 / (((\beta + 4)^2 + 4\gamma) - (2\alpha)^2 / ((\beta + 6)^2 + (2\alpha)^2)) - \dots} \\ & + \frac{(2\alpha)^2}{(\beta - 2)^2 + 4\gamma - (2\alpha)^2 / (((\beta - 4)^2 + 4\gamma) - (2\alpha)^2 / ((\beta - 6)^2 + (2\alpha)^2)) - \dots} \dots \end{aligned} \quad (9)$$

In the interior of the first stability region, β lies strictly between 0 and 1; we shall restrict our analysis to this region. The coefficients D_{2n} are given by similar continued fraction expressions:

$$D_{2n} = -\frac{D_{2n-2}}{2\alpha} \frac{(2\alpha)^2}{(2n + \beta)^2 + 4\gamma - (2\alpha)^2 / (((2n + 2 + \beta)^2 + 4\gamma) - (2\alpha)^2 / ((2n + 4 + \beta)^2 + 4\alpha)) - \dots} \quad (10)$$

and

$$D_{2n} = -\frac{D_{2n+2}}{2\alpha} \frac{(2\alpha)^2}{(2n + \beta)^2 + 4\gamma - (2\alpha)^2 / (((2n - 2 + \beta)^2 + 4\gamma) - (2\alpha)^2 / ((2n - 4 + \beta)^2 + 4\alpha)) - \dots} \quad (11)$$

The technical details involved in finding the continued fractions in (9,10,11) can be found in [11]. In cases of present interest, the coefficients D_{2n} decrease in magnitude rapidly with n . Therefore, we take as an approximation to the solution of (6) a 5-term truncation of (8):

$$z_0 \cong A \sum_{n=-2}^2 D_{2n} \cos((2n + \beta)\hat{t}) + B \sum_{n=-2}^2 D_{2n} \sin((2n + \beta)\hat{t}). \quad (12)$$

For our purposes, keeping five sine and five cosine terms in the expression for z_0 is more than enough. To specify fully the solution at $O(\epsilon^0)$, we must determine the evolution of z_0 on the slow time scale, τ . We do this by letting the amplitudes of the linearly independent parts of the truncated solution (12) vary on the slow time scale:

$$z_0(\hat{t}, \tau) \cong A(\tau) \sum_{n=-2}^2 D_{2n} \cos((2n + \beta)\hat{t}) + B(\tau) \sum_{n=-2}^2 D_{2n} \sin((2n + \beta)\hat{t}). \quad (13)$$

At $O(\epsilon^1)$, the differential equation is the same linear Mathieu equation as at $O(\epsilon^0)$, now driven by functions of z_0 and its derivatives. Driving terms appear in (7) through the contributions of the weak damping, nonlinearity, and secondary excitation. To ensure asymptoticness of the solution up to times of order $1/\epsilon^2$, one normally enforces solvability conditions at $O(\epsilon^1)$ to remove all inhomogeneous terms that cause secular growth of the z_1 solution—assuming these should not be present. In a more typical example, one would solve a sequence of simple oscillator problems with a single natural frequency. In that context, the pair of slowly varying amplitudes A and B are sufficient to remove all secular terms at $O(\epsilon^1)$.

However, the present situation is considerably more complicated. Here we are solving an inhomogeneous Mathieu equation at $O(\epsilon^1)$. Consequently, we must carefully analyze which external forcing frequencies cause secularity. Kotowski [12] shows that a Mathieu equation with homogeneous solution (8) subjected to periodic external forcing with frequency ω_e :

$$\ddot{y} - 4(\gamma + \alpha \cos 2t)y = F \cos \omega_e t \quad (14)$$

has unbounded oscillations only when $\omega_e = 2n + \beta(\alpha, \gamma)$, with β given by (9). These resonant solutions of the externally forced Mathieu equation grow linearly in time.

In the present context, terms on the right-hand-side of (7) play the roles of $F \cos \omega_e t$ in (14) with F depending on the coefficients D_{2n} from z_0 . Each such resonant term causes growth. However, as n increases, the slope of the linearly growing envelope is proportional to D_{2n} , where D_{2n} is the coefficient of the term in the homogeneous solution whose frequency matches the forcing frequency. Thus, although resonances for all n grow linearly in time, those with smaller n are much more dangerous in the sense of destroying asymptoticness of the expansion earlier. In the context of our current perturbation scheme, we therefore choose to enforce solvability conditions which remove only the most dangerous resonant terms from the right-hand-side of (7). Because we have introduced only two slowly varying amplitudes, $A(\tau)$ and $B(\tau)$, we can eliminate only the sine and cosine functions containing a single resonant forcing frequency.

Unfortunately, the truncated series solution (13) introduces forcing terms at multiple resonant frequencies at $O(\epsilon^1)$. As a result, our perturbation scheme loses asymptoticness at first order. But, the central point is that it loses asymptoticness when $z_1 = O(\epsilon^{-1})$. This occurs when the *fastest* growing secular term due to remaining resonant forcing grows to $O(\epsilon^{-1})$. If we make use of the previous scaling arguments for the growth of resonant envelopes for linear Mathieu equations, we find that the expansion should remain asymptotic for times $t = O(t_m)$, where

$$t_m \propto \left(\frac{1}{D_{2n}\epsilon} \right).$$

To make the method as effective as possible, it is sensible to remove the resonant frequency that causes the smallest t_m . Elimination of this frequency from (7) gives two first-order nonlinear differential equations for the two slowly varying amplitudes. We remark that an alternative approach would consist in multiplication of each sine and cosine term of z_0 by an independent, slowly varying amplitude. Then it is possible to remove all secular terms in (7), but (6)—the leading order equation—becomes violated.

2.2. Resonance conditions

It can be observed through numerical experiments that solutions to (2) decay to zero, except for a set of special ω_f values, which we call resonant values. For these exceptional ω_f , steady oscillations with large amplitudes occur. Our goal at this point is to find values of ω_f that cause large-amplitude solutions. The multiple scales method described in Section 2.1 captures both the decaying solutions and the resonant ones. The essential difference between these two types of solutions is their behavior on the long time scale. The multiple scales method reveals that resonances in (2) occur when the special values of ω_f introduce secular terms at $O(\epsilon^1)$ which are absent for arbitrarily chosen ω_f . A more detailed examination of the right-hand-side of (7) reveals the special set of frequencies ω_f that drive resonant behavior. It also helps us choose which rapidly growing secular terms can be eliminated most profitably.

In particular, (7) contains trigonometric terms due to the secondary forcing which have the form:²

$$\begin{aligned} \cos(2\omega_f \hat{t})z_0 &= \cos(2\omega_f \hat{t})A(\tau) \sum_{n=-2}^2 D_{2n} \cos((2n + \beta)\hat{t}) \\ &= \frac{1}{2}A(\tau) \left[\sum_{n=-2}^2 D_{2n} \cos((2n + \beta + 2\omega_f)\hat{t}) + \sum_{n=-2}^2 D_{2n} \cos((2n + \beta - 2\omega_f)\hat{t}) \right]. \end{aligned} \quad (15)$$

These terms contain frequencies that force secular behavior at first order if they match a basic frequency of the homogeneous solution to (7). That is, ω_f can be chosen such that

$$(2n + \beta) \pm 2\omega_f = \pm(2m + \beta),$$

where $2m + \beta$ are the basic frequencies of $M(z_1) = 0$. Upon rearrangement, we find that secondary forcing introduces additional secular terms to (7) through the terms in (15) when ω_f takes the special values:

$$\omega_{f,\text{res}} = p + \beta \quad \text{for } p \in \mathbb{Z}.$$

Similar results are obtained by the authors in [2] using a different method for a linear QP Mathieu equation without damping. They use a regular perturbation method to solve a sequence of Mathieu equations. They solve their, somewhat simpler, $O(\epsilon)$ problem exactly. Their technique provides the same $\omega_{f,\text{res}}$ values, however they report that the method incurs significant error when the secondary frequency is near or at one of these resonant values. Our technique captures the solution at resonance, near resonance, and far from resonance.

When ω_f is set to a nonresonant value, i.e. $\omega_f \neq p + \beta$, removing a secular frequency from the right-hand-side of (7) produces a set of first order differential equations for A and B that determine the evolution of the solution on the slow time scale. When ω_f is set to a resonant value, additional terms must be removed from the right-hand-side of (7). These additional terms modify the differential equations for the slowly varying amplitudes, producing the qualitative change in the solution on the slow time scale that occurs when the system moves from nonresonance ($\omega_f \neq \omega_{f,\text{res}}$) to resonance ($\omega_f = \omega_{f,\text{res}}$). The equations for A and B contain 57 terms at resonance and are too long to show in their entirety. However, they have the relatively simple form:

$$\frac{\partial A}{\partial \tau} = g_1 B^3 + g_2 A^2 B + g_3 A + g_4 B, \quad (16)$$

$$\frac{\partial B}{\partial \tau} = h_1 A^3 + h_2 A B^2 + h_3 B + h_4 A, \quad (17)$$

where the g_i and h_i depend on the coefficients D_{2n} and the parameters in (2).

² For brevity, we show here only the cosine part of z_0 . Similar results are obtained for the sine component.

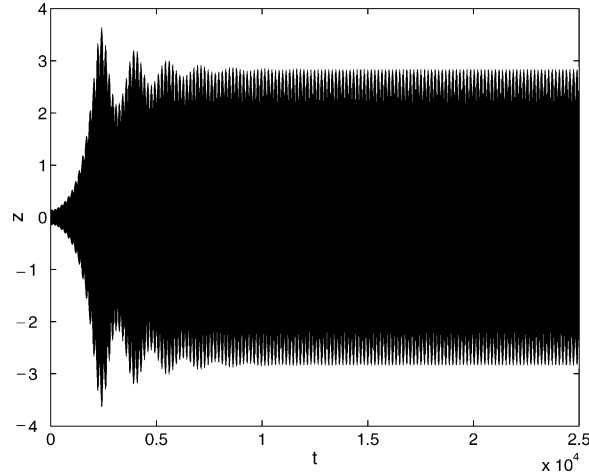


Fig. 1. When $\omega_f = \beta$, there exist oscillations associated with flow on a knotted torus attractor which persists through relatively large increases in damping. This is a large-amplitude oscillation, and we will refer to it as the central resonance. Parameters: $\alpha = 0.15$, $\gamma = -0.05$, $\mu = \delta = \chi = 1$, $\epsilon = 10^{-3}$, $\omega_f = \beta = 0.5094$.

Numerical simulations confirm that solutions for predicted resonant values of ω_f and those for values of ω_f sufficiently far from resonance are qualitatively different. Far from resonance, the dissipation drives solutions to zero exponentially fast, regardless of initial conditions³. At the resonant values predicted by the multiple scales analysis, open sets of initial conditions are attracted to knotted tori. Figs. 1–3 illustrate these distinct types of behavior in the time series. We consider the geometry and topology of the associated attractors in Section 3. We note that while resonant secondary frequencies $\omega_f = p + \beta$ with $p \neq 0$ cause resonance, these higher resonances are more delicate. That is, we must provide strong secondary excitation and weak damping in order to observe them in numerical simulations. Consequently, for the remainder of the paper we will concentrate on the more robust resonance when $\omega_f = \beta$, which we call the central resonance.

2.3. Solutions at central resonance

To apply the method described in Section 2.1, we must choose which secular frequency to eliminate from (7) in order to obtain the solvability conditions that generate evolution equations for the slowly varying amplitudes. This choice must be based on the strength of the secular forcing terms. By strength, we mean that terms causing faster secular growth are stronger than terms causing slower growth. The choice to eliminate the strongest secular terms allows us to ensure that our solution is the best approximation possible (within the scheme) and that it will remain asymptotic for the longest time. With this in mind we note that, when $\omega_f = \beta$, the terms in (7) resulting from the secondary excitation have the form:

$$\begin{aligned} \cos(2\omega_{f,\text{res}}\hat{t}) \sum_{n=-2}^2 D_{2n} \cos((2n + \beta)\hat{t}) &= \frac{1}{2} \sum_{n=-2}^2 D_{2n} [\cos(2n + \beta + 2\omega_{f,\text{res}})\hat{t} + \cos(2n + \beta - 2\omega_{f,\text{res}})\hat{t}] \\ &= \frac{1}{2} \sum_{n=-2}^2 D_{2n} [\cos(2n + 3\beta)\hat{t} + \cos(2n + \beta)\hat{t}]. \end{aligned} \quad (18)$$

³ We consider only initial conditions within the region $R = \{(z, \dot{z}) : |z| < 5, |\dot{z}| < 5\}$ in the $(z, \dot{z})$ -plane. More specifically, we do not analyze the effects of small nonlinearities far away from the origin in phase space.

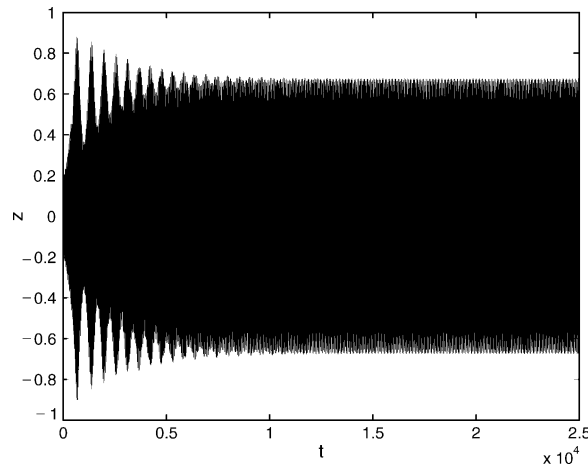


Fig. 2. The secondary frequency is set to a resonant value $\omega_f = 2 + \beta$. The solution is attracted to a knotted torus. The amplitude of the solution is small for this resonant ω_f ; the torus and resulting oscillation is easily destroyed by small increases in damping. Parameters: $\alpha = 0.25$, $\gamma = 0.001$, $\epsilon = 10^{-3}$, $\mu = 0.8$, $\delta = 10$, $\chi = 5$, $\omega_f = 2 + \beta = 2.3674$.

The dangerous frequencies in this expression are the $2n + \beta$. Because they match the frequencies appearing in the homogeneous solution to the Mathieu equation, they produce secular growth in the solution for z_1 at $O(\epsilon^1)$. The envelope associated with each of these secular terms grows like $D_{2n}\hat{t}$. The D_{2n} is the coefficient of the term in the homogeneous solution whose frequency matches that of the secular driving term with frequency $2n + \beta$. All of the terms on the right-hand-side of (7) introduce similar secular forcing. Removal of a particular driving frequency leaves behind other secular terms whose growth rates must be considered to determine the period of time over which the solution remains asymptotic. Strong growth at $O(\epsilon^1)$ reduces the time over which the solution is asymptotic and rapidly introduces error leading to a poor approximation.

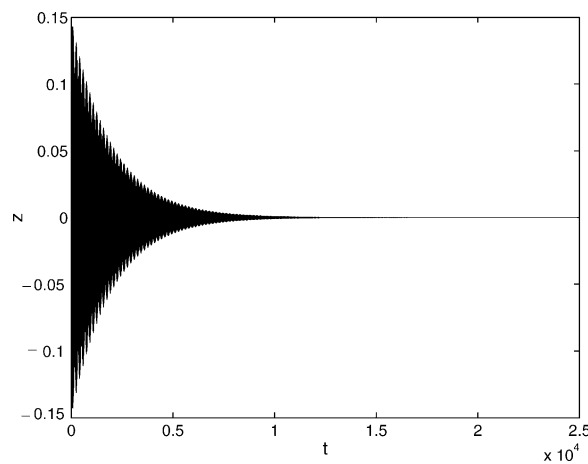


Fig. 3. Solutions decay for ω_f sufficiently far from a resonant value. Parameters: $\alpha = 0.15$, $\gamma = -0.05$, $\mu = \delta = \chi = 1$, $\epsilon = 10^{-3}$, $\omega_f = 2\beta = 1.0187$.

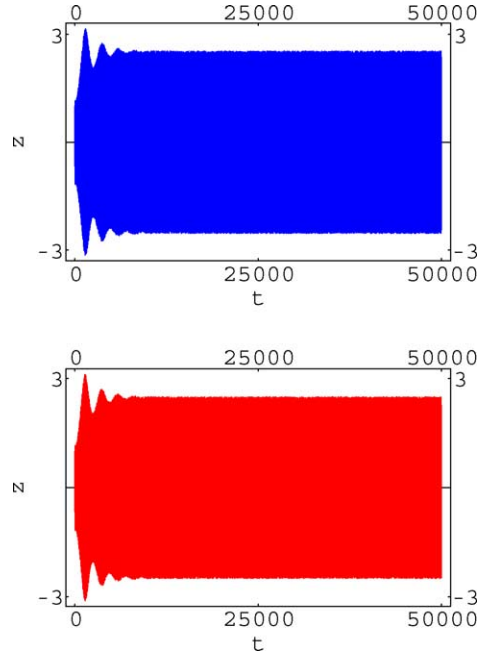


Fig. 4. At favorable values of α and γ that make D_{-2} small (-0.07), the perturbation approximation (top) and the numerical solution (bottom) show the same behavior on the slow time scale. When overlaid, the plots of the two solutions are indistinguishable. Fig. 5 reveals the solution behavior on the fast time scale. Similar results are obtained for parameters which make D_{-2} smaller. Parameters: $\alpha = 0.05$, $\gamma = -0.1$, $\mu = \delta = \chi = 1$, $\epsilon = 10^{-3}$, $\omega_f = \beta$.

As noted earlier, the D_{2n} decrease in magnitude rapidly for increasing n . Removal of the secular driving terms with frequency β eliminates the fastest growing secular terms which grow in proportion to D_0 . This leaves a variety of secular terms, the strongest of which grows in proportion to D_{-2} . Thus, the method should remain asymptotic for times $t_m = O(1/D_{-2}\epsilon)$. In practice, this requires that α and γ be chosen to make $|D_{-2}|$ small in order to ensure asymptoticity for sufficiently long times.

We show in Figs. 4 and 5 both numerical solutions and perturbation approximations at the central resonance for an (α, γ) pair that makes $D_{-2} = -0.07$. The perturbation approximation captures the high-frequency oscillation on the fast time scale, as well as the slow evolution of the envelope. The dynamics of the system of slowly varying amplitude equations is best understood by examining slow-time solution trajectories in the (A, B) -plane. At resonance, the trajectories in this space spiral into a stable, attracting fixed point (Fig. 6). The presence of this stable focus together with the complicated solution on the fast time scale leads to knotted torus attractors whose topology will be discussed below.

For comparison, we also show here perturbation approximations and numerical solutions (see Figs. 7–9) for an (α, γ) pair that makes $D_{-2} = -0.11$, an increase from the previous case. This results in a less accurate approximation to the solution. The approximation still captures the qualitative behavior and also reveals the approach to the attractor for long times. It also captures the high frequency “ripple” behavior on the fast time scale. However, the increased strength of remaining secular terms introduces error. As can be seen in Fig. 7, the perturbation approximation slightly overestimates the amplitude of the solution envelope. The errors are larger for the transient stage of the solution (Fig. 8), but are decreased at steady state (Fig. 9). The method still correctly predicts the presence of an attractor due to a stable fixed point in the (A, B) -plane.

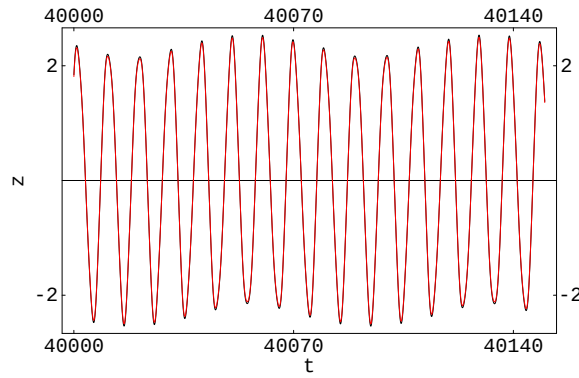


Fig. 5. Both the perturbation approximation and numerical solution are plotted in this figure. The two solutions lie on top of one another demonstrating the close agreement between the approximate and numerical results. Parameters are those of Fig. 4.

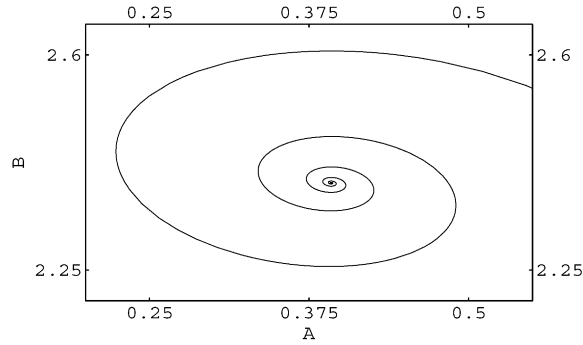


Fig. 6. The perturbation approximation for the slowly varying amplitudes spirals into a fixed point at resonance. Parameters are those of Fig. 4.

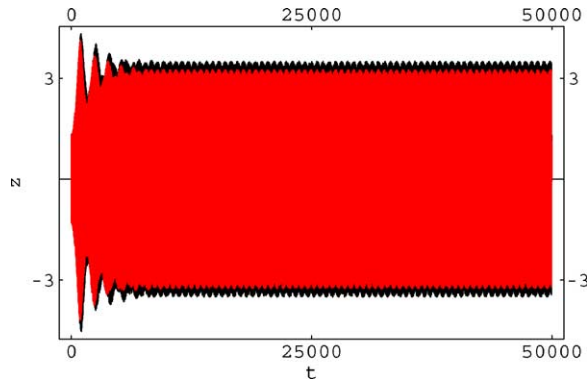


Fig. 7. At a less favorable combination of α and γ that leads to larger D_{-2} (-0.11), the larger amplitudes of remaining secular terms lead to increased error in the perturbation approximation (dark) as compared to the numerical solution (light). Parameters: $\alpha = -0.04$, $\gamma = 0.125$, $\mu = \delta = \chi = 1$, $\epsilon = 10^{-3}$, $\omega_f = \beta$.

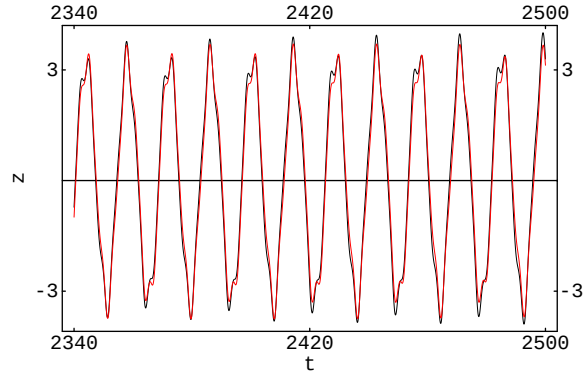


Fig. 8. Comparison of the perturbation approximation (dark) and the numerical solution (light) before transients have decayed. Parameters are those of Fig. 7.

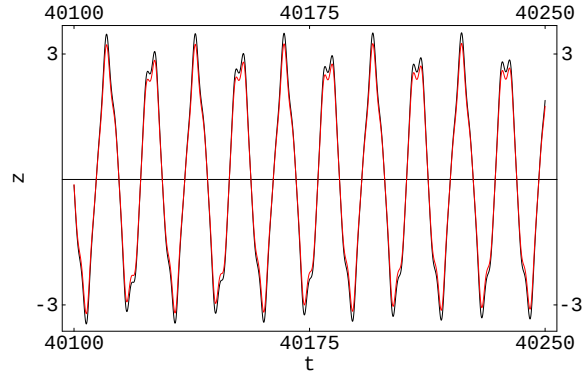


Fig. 9. Comparison of the perturbation approximation (dark) and the numerical solution (light) after the decay of transients. Parameters are those of Fig. 7.

3. Bifurcations and topology of solutions

3.1. Solutions near the central resonance

Both examples in Section 2.3 (Figs. 4–6 and 7–9, respectively) demonstrate that steady, large amplitude solutions can occur. While the approximation scheme yields both the resonant frequencies and the resonant solutions, it remains to be shown *how* the attractor is engendered. The method of multiple scales, as constructed, does not give information about the bifurcations that lead to the birth and death of the large-amplitude attractor. Additional insight about the bifurcations leading to resonant solutions can be gained by studying the behavior in an interval of secondary frequencies that surrounds the resonant frequency. This can be done by including detuning in the multiple scales scheme. We define the secondary frequency of (2) to have the form:

$$\omega_f = \beta + \epsilon\nu. \quad (19)$$

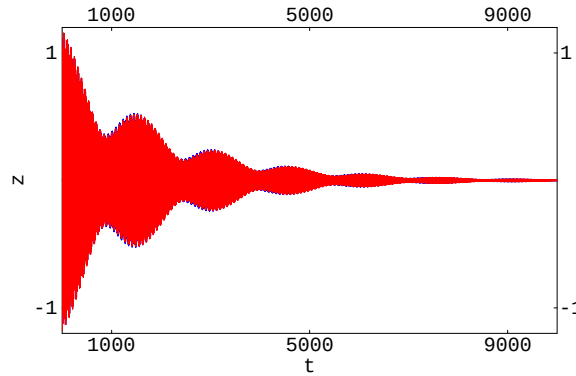


Fig. 10. When detuned, the perturbation approximation and the numerical solution decay for secondary forcing frequencies sufficiently far away from the resonant frequency. Here, $\omega_f = \beta - 2.6\epsilon$. Other parameters are those of Fig. 4.

Thus, for $\nu \neq 0$ the secondary forcing is shifted slightly away from the resonant frequency. In the context of the multiple scales analysis, detuning is equivalent to the introduction of forcing on the $\tau = \epsilon\hat{t}$ time scale:

$$\cos(2\omega_f t) = \cos(2\beta t + 2\epsilon\nu t) = \cos(2\beta\hat{t} + 2\nu\tau).$$

The solution on the fast time scale is still the solution to the $O(\epsilon^0)$ Mathieu equation, and the evolution equations on the slow time scale are still given by enforcing solvability conditions at $O(\epsilon^1)$. The key change is that the slow time evolution equations are now nonautonomous in the case of detuning. The nearness to resonance, measured by ν , appears in the evolution equations for the slowly varying amplitudes as forcing on the τ time scale with frequency 2ν . Study of the behavior of the dynamical system near resonance then reduces to study of the (now nonautonomous) system for the slowly varying amplitudes over a range of ν .

For sufficiently negative values of ν , the origin is found to be stable and attracting, in contrast to the situation at resonance. The numerical solution and perturbation approximation agree on both the slow time scale (Fig. 10) and on the fast time scale (Fig. 11). Because the slow-time evolution equations are periodically forced due to detuning, it is convenient to observe the solution using a Poincaré map. In this simple case, the map has a stable fixed point at the origin (not shown).

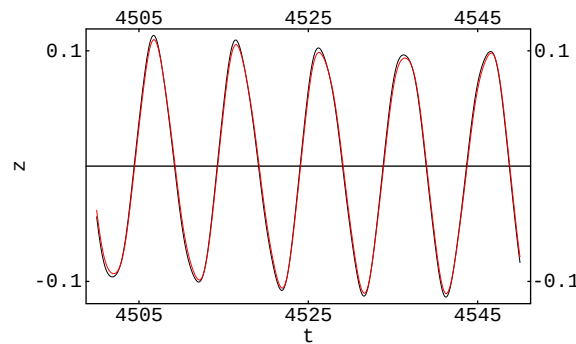


Fig. 11. The perturbation approximation agrees well with numerical solutions on both the fast time scale (seen here) and on the slow time scale (see Fig. 10). Parameters are those of Fig. 10.

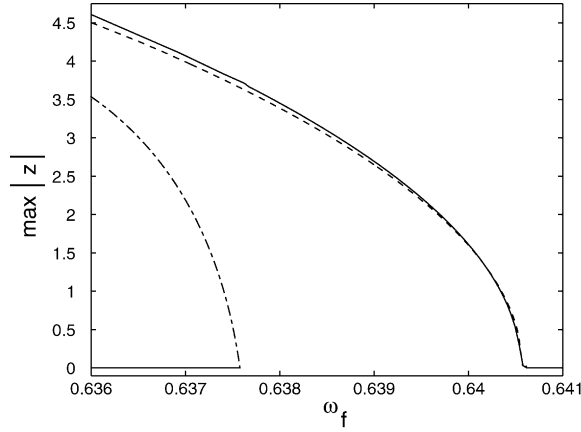


Fig. 12. The stable branches of the response diagram produced by the method of multiple scales (solid lines) closely approximate those generated numerically (dashed lines). An unstable branch (dash-dot line), shown schematically, separates the stable branches. The approximation technique captures the amplitude and stability type of solutions, as well as the location of bifurcations in the parameter space. The parameters: $\alpha = 0.05$, $\gamma = -0.1$, $\mu = \delta = \chi = 1$, $\epsilon = 10^{-3}$.

By use of the detuning scheme, we now study the system over a range of secondary forcing frequencies by varying ν . We show in Fig. 12 a response diagram constructed by plotting $\|z\|_\infty$ versus ω_f . The dynamics on the two horizontal branches are similar to the dynamics shown in Fig. 10 and are associated with the decay of trajectories to the trivial solution. The dynamics on the large-amplitude stable branch are more interesting. Here, we observe steady, large amplitude oscillations similar to those exactly at the resonance (Figs. 13 and 14). The principal difference between solutions on the trivial branches and solutions on the large-amplitude branch lies in the slow-time dynamics. On the large-amplitude branch, the solution for the slowly varying amplitudes becomes entrained in a limit cycle that is 2-periodic in $\tau = \epsilon t$ with the frequency 2ν driving associated with the detuning (Fig. 15). Thus, the attractor is covered by a curve that is approximated by the product of the 2-periodic functions $A(\tau)$ and $B(\tau)$ and the solution to the homogeneous Mathieu equation (13).

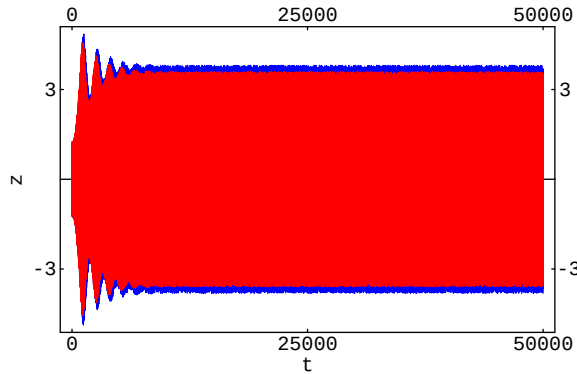


Fig. 13. The steady oscillation present at resonance persists over a small interval of ω_f which contains the exact resonant value. Here, $\omega_f = \beta - 0.5\epsilon$, and both the perturbation approximation and numerical solution are attracted to a knotted torus at this value. Parameters: $\alpha = 0.05$, $\gamma = -0.1$, $\mu = \delta = \chi = 1$, $\epsilon = 10^{-3}$.

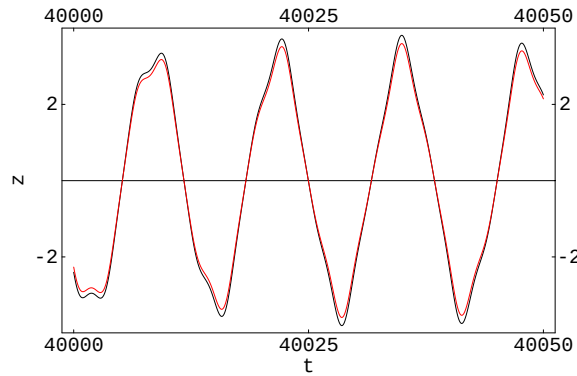


Fig. 14. The detuned perturbation approximation agrees well with numerical solutions on both the fast time scale (seen here) and on the slow time scale (see Fig. 13). Parameters are those of Fig. 13.

3.2. Topological bifurcations of resonant solutions

In the response diagram in Fig. 12, two bifurcations are implicated in the generation of the resonant peak. They occur for ω_f near 0.6375 and 0.6405. At the lower frequency, an unstable branch (not shown) bifurcates from the stable trivial branch, which becomes unstable thereafter. As ω_f is increased across the higher frequency, the stable large-amplitude branch meets the unstable trivial solution and stabilizes it. The behavior of the associated attractors indicated by the response diagram is best understood using a topological approach.

We first address the two-frequency excitation in a way that is amenable to topological analysis. To do so, we suspend the nonautonomous equation (2) in $\mathbb{R}^2 \times T^2$ by introducing an additional phase space variable for each cyclic forcing term. The equation becomes

$$\dot{q}_1 = q_2, \quad \dot{q}_2 = -\mu q_2 - 4(\gamma + \alpha \cos \theta_1 - \epsilon \cos \theta_2)(-q_1 + \beta q_1^3), \quad \dot{\theta}_1 = 2, \quad \dot{\theta}_2 = 2\omega_f.$$

The suspended system generates a flow that is a map from $\mathbb{R}^2 \times T^2$ to itself.

Now, we turn to a consideration of Poincaré maps for the system. Rather than examining a bi-infinite sequence of planar maps [13], we shall consider the action of two, three-dimensional maps. This idea is developed in our

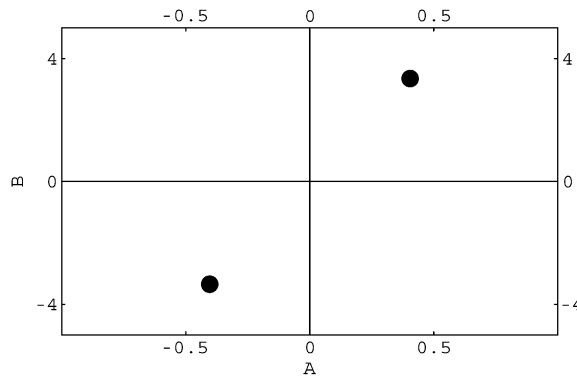


Fig. 15. The Poincaré section for the slow-time evolution equations shows a 2-periodic response after the decay of transients. Parameters are those of Fig. 13.

previous paper [8]. Such a viewpoint gives a more detailed understanding of the geometry by allowing us to consider the iterated action of the maps on the same spaces. We define two essentially orthogonal, global, three-dimensional Poincaré sections of the flow

$$\Sigma^{\theta_{10}} = \{(q, \theta) \in \mathbb{R}^2 \times T^2 \mid \theta_1 = \theta_{10}\}, \quad \Sigma^{\theta_{20}} = \{(q, \theta) \in \mathbb{R}^2 \times T^2 \mid \theta_2 = \theta_{20}\}.$$

The associated Poincaré maps on the sections are

$$P_1 : \Sigma^{\theta_{10}} \rightarrow \Sigma^{\theta_{10}}$$

$$(q_1(0), q_2(0), \theta_{10}, \theta_2(0)) \mapsto (q_1(\pi), q_2(\pi), \theta_{10}, \theta_2 + 2\pi\omega_f)$$

and

$$P_2 : \Sigma^{\theta_{20}} \rightarrow \Sigma^{\theta_{20}}$$

$$(q_1(0), q_2(0), \theta_1(0), \theta_{20}) \mapsto \left(q_1\left(\frac{\pi}{\omega_f}\right), q_2\left(\frac{\pi}{\omega_f}\right), \theta_1 + \frac{2\pi}{\omega_f}, \theta_{20} \right).$$

The orbits of the maps P_1 and P_2 lie in $\Sigma^{\theta_{10}} = \mathbb{R}^2 \times \{\theta_{10}\} \times S^1$ and $\Sigma^{\theta_{20}} = \mathbb{R}^2 \times S^1 \times \{\theta_{20}\}$, respectively. Because the vector field generating the flow is C^∞ , the flow is a diffeomorphism on $\mathbb{R}^2 \times T^2$. Consequently, the P_i are also diffeomorphisms on the sections $\Sigma^{\theta_{i0}}$.

In [8] we show that the change in knot type at a TTB of knotted torus attractors in the full phase space is captured by this pair of maps. Such bifurcations can be observed by examining the topology of invariant 1-manifolds that exist in the Poincaré sections defined above. These 1-manifolds have the form of (closed) braids. For the present purposes, we need only recall that the number of strands in a braid is equal to the degree of the torus attractor, and the degree is an isotopy invariant which can change only at a bifurcation.

With this in mind, we first consider the braids present in the Poincaré sections shown in Fig. 16 for solutions on the stable portions of the trivial branch in the response diagram. Here, the decay to the trivial solution is reflected in the transient approach to the long term attractor in the cross-sections. The simple topology of the attractor, a single strand braid, is faithfully reproduced by the multiple scales approximation. For initial conditions leading to solutions on the large-amplitude stable branch, trajectories approach the attractor shown in the Poincaré sections of Fig. 17. The braids associated with this attractor are both 2-strand braids. As mentioned before, the number of strands is equal to the degree of the surface and is isotopy invariant. Thus, this torus attractor in the full phase space $\mathbb{R}^2 \times T^2$ has different knot type from the torus associated with the single strand braid. These torus attractors of different knot type are introduced to and eliminated from the phase space through TTBs.

The bifurcation at $\omega_f \simeq 0.6405$ is a type-II TTB of the doubling variety [8]. By this we mean that as ω_f is reduced across the bifurcation value, the trivial single strand braids in the two Poincaré sections on the trivial branch lose stability and shed stable two strand braids. In the section $\Sigma^{\theta_{10}}$ the number of components (one) remains constant while the number of components doubles to two in $\Sigma^{\theta_{20}}$. This topological transition can be tracked by a series of numerical simulations and is also captured by the detuned multiple scales approximation. The TTB appears in the multiple scales approximation as a period-doubling in the slow-time evolution equations (Fig. 18). The bifurcation at $\omega_f \simeq 0.6375$ is a similar bifurcation with the stability roles reversed. It is also captured by the multiple scales calculations.

The approximation scheme also makes it substantially easier to locate knotted invariant tori. This can be done using the two-dimensional Poincaré map, P_s , for the non-autonomous (A, B) system. The location of fixed points in P_s^2 , the composition of P_s with itself, determines the initial conditions at which trajectories on the tori are anchored in the flow. More specifically, a point in the domain of P_s^2 located at (A_0, B_0) in the (A, B) -plane corresponds to initial conditions $(z_0, \dot{z}_0)$ for (2). Geometrically, this indicates that the (A, B) -plane is equivalent to the manifold $\mathbb{R}^2 \times \{0\} \times \{0\}$. For convenience, we choose $\theta_{10} = 0$ and $\theta_{20} = 0$ when constructing the three-dimensional Poincaré

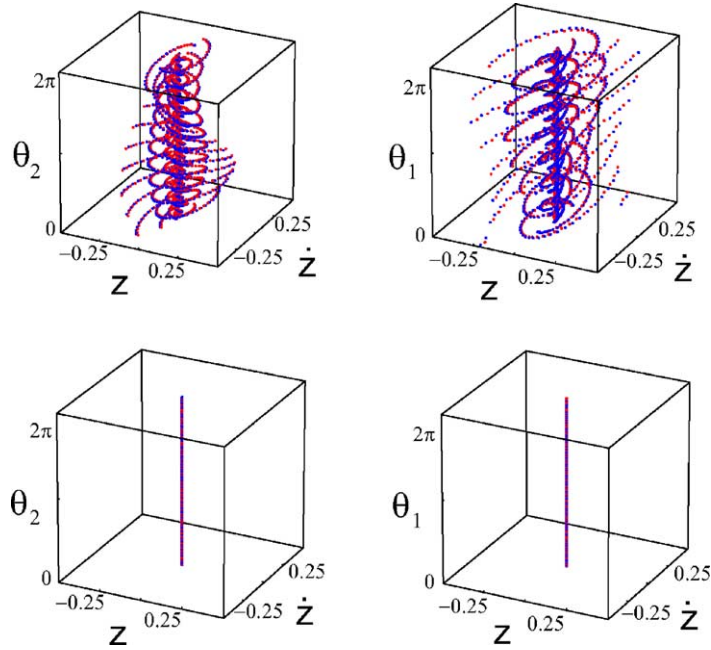


Fig. 16. Nonresonant solutions decay to the trivial braid on one strand that lies along the origin of the $(z, \dot{z})$ -plane in the Poincaré sections $\Sigma^{\theta_{10}}$ (left) and $\Sigma^{\theta_{20}}$ (right). The top figures include the transient approach for both the numerical solution (light) and the perturbation approximation (dark). The bottom figures depict the attractor cross-sections after the decay of transients. Parameters are those of Fig. 10.

sections for the full system. Then, the $\theta_i = 0$ plane in each of the two Poincaré sections (the “bottom” plane) is equivalent to $\mathbb{R}^2 \times \{0\} \times \{0\}$. Thus, fixed points in the (A, B) -plane correspond to the bases of strands of braids in $\Sigma^{\theta_{10}}$ and $\Sigma^{\theta_{20}}$, and consequently are also points lying in invariant tori. For the case of incommensurate forcing frequencies, an initial condition anchored at this point generates a solution that densely covers the invariant torus. This is particularly helpful for locating saddle-type tori. While these objects are difficult to find by numerical integration, it is relatively straightforward to find fixed points of P_s^2 , regardless of stability type.

The multiple scales technique also makes accessible the stable and unstable invariant manifolds of the saddle-type tori (Fig. 18). These can be found by tracing out the stable manifolds of the saddle-type fixed points of P_s^2 . By an argument similar to that just presented, the set of points in the stable manifold of a saddle-type fixed point of

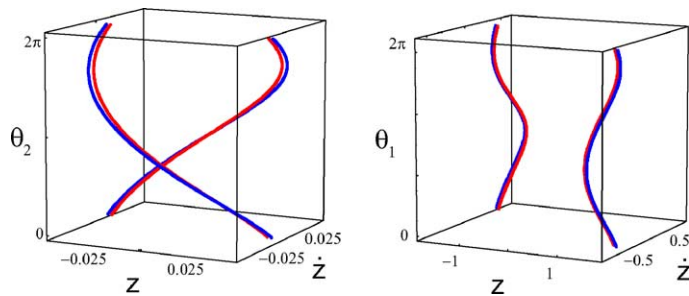


Fig. 17. Resonant solutions are attracted to nontrivially knotted tori whose cross-sections are braids on two strands. Thus, resonant attractors are topologically distinct from nonresonant attractors. The resonant attractors are introduced to and eliminated from the phase space by TTBs. Parameters are those of Fig. 13.

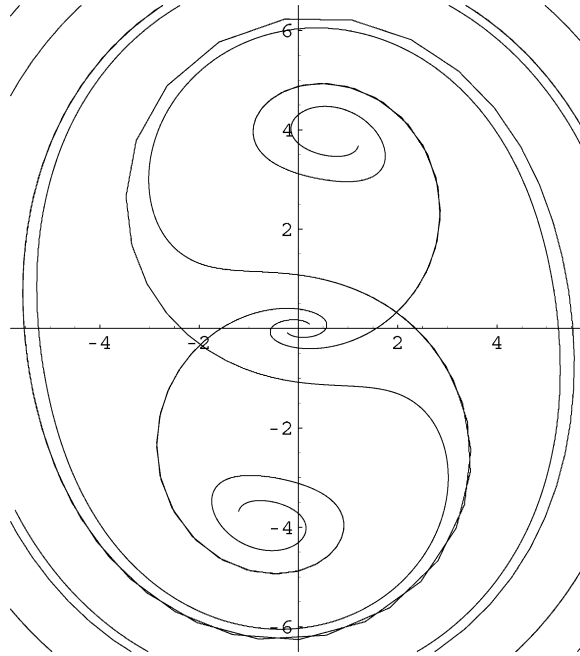


Fig. 18. Shown here in the (A, B) -plane are the stable and unstable invariant manifolds of the saddle-type fixed points of $(P_s)^2$ for $\omega_f = 0.637$ (see Fig. 12). If ω_f is increased sufficiently, the saddle points will collapse on the origin to destabilize it. Further increasing ω_f will force the two sinks to the origin to stabilize it. This pair of bifurcation in the map corresponds to TTBs in the four-dimensional phase space. Parameters: $\alpha = 0.05$, $\gamma = -0.1$, $\mu = \delta = \chi = 1$, $\epsilon = 10^{-3}$.

P_s^2 represents a slice of the invariant 2-manifold of the saddle-type braids in the Poincaré sections $\Sigma^{\theta_{10}}$ and $\Sigma^{\theta_{20}}$, and thus corresponds to a curve of points in the invariant 3-manifold of the saddle-type invariant torus in $\mathbb{R}^2 \times T^2$. These three-dimensional invariant manifolds are codimension 1 and divide the phase space into disjoint basins of attraction for the stable knotted tori. Thus, by mapping out the basins of attraction for P_s^2 , we can determine which knotted torus in $\mathbb{R}^2 \times T^2$ is approached by a particular set of initial conditions.

4. Conclusions

The damped, nonlinear Mathieu equation with two-frequency parametric forcing supports large-amplitude resonant oscillations for appropriately chosen secondary forcing frequencies, ω_f . A variant of the method of multiple scales reveals the resonant values of ω_f and approximations to resonant solutions. The introduction of detuning enables the capture of a pair of period doubling bifurcations in the slow-time equations which generate the resonant peak in the response diagram. By adopting a topological approach, it is shown that these bifurcations are type-II TTBs of the doubling variety, that we described in an earlier work [8]. The detuned perturbation approximations near resonance capture the topology of solutions in the $\mathbb{R}^2 \times T^2$ phase space. They provide a closed form description of trajectories on knotted 2-tori in the four-dimensional phase space. Finally, analysis of the slow-time Poincaré map composed with itself provides approximations for the location of saddle-type invariant tori and their stable invariant 3-manifolds which form basin boundaries in $\mathbb{R}^2 \times T^2$ and organize the phase space.

These results help clarify the nature of solutions which densely cover knotted tori. That is, flow on a knotted torus near resonance is characterized by the presence of multiple frequencies, including the several frequencies from the

fast dynamics and the frequency of the slow, periodic amplitudes. While solutions with many frequencies could not be captured by a traditional multiple scales routine that admits only single-frequency fast dynamics, our technique captures the multiple frequencies and the associated topology. It also makes clear that there is a fundamental relation between the slow dynamics, the fast dynamics, and the knot type of the resulting attractor. However, it is worth emphasizing that our work here has not elucidated the exact nature of this relation. For example, why does a pair of type II TTBs occur, as opposed to a pair of type I? Is the number of components in the sections $\Sigma^{\theta_{i0}}$ controlled by the fast dynamics, the slow dynamics, or an interaction between the two? The techniques developed here should help to answer these and related questions in future investigations.

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